Rectifier System Inspection, Preventive Maintenance & Load Testing Checklist

Region		
RTOM		
Station		
Rectifier Make/Model		
Capacity		
Date		
Checked by	Name:	Signature:

Maintenance Schedule

- Quarterly preventive maintenance (4 services per unit per year)
- Monthly operational inspections (12 inspections per unit per year)

A. Routine Visual Inspection Checklist

No.	Inspection Item	Status	Remarks
A1	Rectifier room is clean and free from dust, water leaks, and obstructions	□ OK □ NOT OK	
A2	Adequate ventilation available inside rectifier room	□ OK □ NOT OK	
A3	Ambient temperature within acceptable limits	□ OK □ NOT OK	
A4	No unusual noise from rectifier modules	□ OK □ NOT OK	
A5	No unusual smell, overheating, or burning marks	□ OK □ NOT OK	
A6	Rectifier cabinet doors and locks are in good condition	□ OK □ NOT OK	
A7	All warning labels and identification tags are available and readable	□ OK □ NOT OK	
A8	No signs of corrosion on cabinet, terminals, or busbars	□ OK □ NOT OK	
A9	No loose, damaged, or exposed cables	□ OK □ NOT OK	
A10	Cable entry points properly sealed	□ OK □ NOT OK	
A11	Earthing conductor securely connected	□ OK □ NOT OK	
A12	DC distribution breakers/fuses properly labeled	□ OK □ NOT OK	

A13	System LEDs and indicators operating normally	□ OK □ NOT OK	
A14	No active alarms displayed on controller	□ OK □ NOT OK	
A15	Rectifier controller display functioning properly	□ OK □ NOT OK	

B. Electrical Inspection Checklist

No.	Inspection Item	Status	Remarks
B1	AC input voltage within specified range	□ OK □ NOT OK	
B2	AC input phase sequence correct (if applicable)	□ OK □ NOT OK	
B3	AC input breakers in good condition	□ OK □ NOT OK	
B4	Surge protection devices healthy	□ OK □ NOT OK	
B5	Rectifier output voltage within set value	□ OK □ NOT OK	
B6	Output current balanced among rectifier modules	□ OK □ NOT OK	
B7	Battery float voltage correct	□ OK □ NOT OK	
B8	Load current indication accurate	□ OK □ NOT OK	
B9	Battery current indication accurate	□ OK □ NOT OK	
B10	Earth continuity satisfactory	□ OK □ NOT OK	
B11	No evidence of hot spots at busbars and terminals	□ OK □ NOT OK	
B12	DC breakers/fuses free from damage and overheating	□ OK □ NOT OK	

C. Preventive Maintenance Checklist

No.	Maintenance Item	Status	Remarks
C1	External surfaces cleaned	□ OK □ NOT OK	
C2	Internal cabinet cleaned using approved method	□ OK □ NOT OK	
C3	Cooling fans checked and cleaned	□ OK □ NOT OK	
C4	Air inlets cleaned/replaced (if applicable)	□ OK □ NOT OK	
C5	Ventilation openings clean and unobstructed	□ OK □ NOT OK	
C6	All AC power connections tightened	□ OK □ NOT OK	
C7	All DC power connections tightened	□ OK □ NOT OK	
C8	Battery connections tightened	□ OK □ NOT OK	
C9	Earthing connections inspected and tightened	□ OK □ NOT OK	
C10	Rectifier modules securely installed	□ OK □ NOT OK	
C11	Module indicators functioning correctly	□ OK □ NOT OK	
C12	Controller firmware/settings verified	□ OK □ NOT OK	

C13	Alarm history reviewed	□ OK □ NOT OK	
C14	Alarm thresholds verified	□ OK □ NOT OK	
C15	AC failure alarm tested	□ OK □ NOT OK	
C16	Rectifier module failure alarm tested	□ OK □ NOT OK	
C17	High DC voltage alarm tested	□ OK □ NOT OK	
C18	Low DC voltage alarm tested	□ OK □ NOT OK	
C19	Battery low-voltage alarm tested	□ OK □ NOT OK	
C20	Remote monitoring interface operational	□ OK □ NOT OK	

D. Battery & DC System Inspection Checklist

No.	Inspection Item	Status	Remarks
D1	Battery bank physically clean	□ OK □ NOT OK	
D2	No battery swelling or deformation	□ OK □ NOT OK	
D3	No electrolyte leakage observed	□ OK □ NOT OK	
D4	Battery terminal connections secure	□ OK □ NOT OK	
D5	Battery temperature within normal range	□ OK □ NOT OK	
D6	Battery room ventilation adequate	□ OK □ NOT OK	
D7	Battery strings properly identified	□ OK □ NOT OK	
D8	Battery voltage within acceptable range	□ OK □ NOT OK	
D9	Inter-cell connectors free from corrosion	□ OK □ NOT OK	
D10	Battery protection devices healthy	□ OK □ NOT OK	

E. Load Testing Checklist

No.	Test Item	Status	Remarks
E1	Load test permit/authorization obtained	☐ OK ☐ NOT OK	
E2	Test equipment calibrated and available	☐ OK ☐ NOT OK	
E3	Battery backup available before testing	☐ OK ☐ NOT OK	
E4	Safety PPE used during testing	☐ OK ☐ NOT OK	
E5	Rectifier operating normally before test	☐ OK ☐ NOT OK	
E6	Load bank connections secure	☐ OK ☐ NOT OK	
E7	Rectifier sustained 25% load without alarms	☐ OK ☐ NOT OK	
E8	Rectifier sustained 50% load without alarms	☐ OK ☐ NOT OK	
E9	Rectifier sustained 75% load without alarms	☐ OK ☐ NOT OK	
E10	Rectifier sustained 100% load without alarms	☐ OK ☐ NOT OK	
E11	Output voltage remained within specification during test	☐ OK ☐ NOT OK	
E12	Output current remained stable during test	☐ OK ☐ NOT OK	
E13	Current sharing between modules satisfactory	☐ OK ☐ NOT OK	

E14	No module overheating during load test	☐ OK ☐ NOT OK	
E15	Cooling fans operated correctly during load test	☐ OK ☐ NOT OK	
E16	No abnormal alarms generated during load test	☐ OK ☐ NOT OK	
E17	Load removed successfully without interruption	☐ OK ☐ NOT OK	
E18	Rectifier returned to normal operation after testing	☐ OK ☐ NOT OK	